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Systems, apparatus, and methods for indicating spectrum sharing

Abstract

The present disclosure relates to systems and methods for indicating to a spectrum access system communication preferences of a base station and/or of a network operator. Different methods may be used to indicate the communication preferences, including a dedicated type parameter transmitted with spectrum allocation requests from the base station, a rule-based system where the spectrum access system accesses communication preferences of the base station and/or network operator through an identifier corresponding to a rule definition, or the like. The spectrum access system may assign a channel to the network operator based on its communication preferences, such as to determine a channel assignment to correspond to a channel aligned with a raster of communications to be used by the network operator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional application claiming priority to U.S. Provisional Application No. 63/041,648, entitled “SYSTEMS, APPARATUS, AND METHODS FOR INDICATING SPECTRUM SHARING,” filed Jun. 19, 2020, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) The present disclosure relates generally to electronic devices, and more particularly, to electronic devices that utilize radio frequency signals, transmitters, and receivers for wireless communication.

(2) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(3) Wireless communication systems are rapidly growing in usage. Wireless electronic devices, such as smartphones and tablet computers, are becoming increasingly sophisticated. In addition to supporting telephone calls, many wireless electronic devices now provide access to the internet, email, text messaging, and navigation using the global positioning system (GPS), and are capable of operating sophisticated applications that utilize these functionalities.

(4) Transmitters and/or receivers may be included in various electronic devices to enable communication between user equipment (e.g., user electronic devices, transmitting or receiving electronic devices, user devices) via wireless networks. The networks may be deployed through a variety of technologies including, but not limited to, access network base stations, such as an eNodeB (eNB) for long-term evolution (LTE) access networks and/or a next generation NodeB (gNB) for 5th generation (5G) access networks. In some electronic devices, a transmitter and a receiver are combined to form a transceiver. Transceivers may transmit and/or receive wireless signals by way of an antenna coupled to the transceiver, such as radio frequency (RF) signals indicative of data.

(5) With the introduction of frequency range 2 (FR2), which includes frequency bands from 24.25 Gigahertz (GHz) to 52.6 GHz, in the release (Rel-16) of the New Radio standard release relating to 5G communications, a network deployment may experience challenges related to sharing and/or utilizing spectrums of the network. In a relatively simple deployment, a network operator managing a network deployed by network access nodes (e.g., radio access nodes, base stations) may be allocated a fixed-width portion from a shared pool of bandwidth (e.g., shared pool of radio spectrum resources). The shared pool of bandwidth may be managed by a spectrum access system (SAS) entity in communication with many network operators, and each network operator may at some point in time desire a respective portion of the shared pool of bandwidth to perform operations of their respective managed networks. However, allocating fixed-width portions of the spectrum may not enable efficient utilization of network resources, and may lead to some managed networks being under-loaded or over-loaded, such as when channel bandwidths are not allocated based on load usages or expected usages.

(6) Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

SUMMARY

(7) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(8) Recent advances in radio frequency network deployments have led to development of a managerial entity that manages allocation of portions of a shared radio spectrum between network operators. For example, recently developed communication standards provide for a shared (e.g., common) pool of bandwidth of a radio spectrum shared between multiple network operators, such as a Citizens Broadband Radio Service (CBRS) band. In this way, a managerial entity, such as a Spectrum Access System (SAS) entity, associated with management of the shared pool of bandwidth, may allocate available portions of the radio spectrum between the network operators. These network operators may include large and/or corporate entities, cellular providers, wireless service providers, wireless carriers, cellular companies, or mobile network carriers, or the like, and the SAS entity may be provided by a regulatory body, governmental body, an oversight corporate

entity, or the like.

(9) The shared pool of bandwidth may be divided between requesting network operators as fixed-width spectrum allocations. However, as described above, fixed-width allocations of a radio spectrum may not enable efficient utilization of network resources, and may lead to some managed networks being under-loaded or over-loaded, such as when allocated channel bandwidths are not allocated based on load usages and/or expected usages.

(10) Dynamic bandwidth allocations may enable utilization of the radio spectrum based on dynamic allocation of resources. In this way, the SAS entity may allocate resources of the radio spectrum between requesting network operators based on a current load and/or expected network loads, thereby permitting efficient allocation of resources by assigning resources based on greatest determined demand. In this way, as loads of access networks increase or decrease, the SAS entity may adjust allocations of the spectrum.

(11) In some cases, requests from the network operator may include an indication of expected load or current load, and/or may indicate an amount of a requested bandwidth determined by the network operator based on changes in expected network loads. Additionally or alternatively, the SAS entity may allocate portions of the radio spectrum based on geographical locations of the network operators and/or of radio access nodes of the network operators, incumbent services expected to be provided by each network operator (e.g., as provided by indicators of future demand and/or current demand for bandwidth), assigned priorities of each network operator, or the like.

(12) Although sharing of the shared pool of resources may lead to more efficient allocations of resources since resources are allocated based on usage and/or a priority of network operators as opposed to fixed allocations, these above-described methods of spectrum allocation may be problematic when dynamic spectrum sharing (DSS) operations are implemented by a network operator. For example, some network operators may communicate exclusively over a first network (e.g., 4G/LTE), some exclusively over a second network (e.g., 5G/NR), while some network operators using DSS operations may use two or more networks to communicate, and thus may benefit from sharing a spectrum between two or more networks. To share a spectrum between multiple network types, such as between a 4th Generation (4G) network or Long Term Evolution (LTE) network, a 5th Generation (5G) or New Radio (NR) network, or the like, the different configurations of each network should be considered and satisfied. The first network may use a first raster alignment (e.g., 100 kilohertz (kHz)) while the second network may use a second raster alignment (e.g., 15 kHz), and the first raster alignment may not be a multiple of the second raster alignment, and thus a third raster alignment (e.g., 300 kHz) representative of a multiple of one or more of the raster alignments may be implemented to compensate raster alignments of both networks.

(13) DSS operations may permit the network operator using a base station to dynamically share portions of the radio spectrum between communications occurring over the first network and communications occurring over the second network, leading to improved levels of spectrum utilization. Levels of spectrum utilization may improve since the network operator is provided one spectrum allocation to be shared between two networks as opposed to two spectrum allocations for two networks. However, as described above, a center frequency and/or raster should be compatible with both the first network and the second network before the network operator is able to share the spectrum allocation between the two networks. Thus, the SAS entity assigning portions of a spectrum may inadvertently prevent an operator from communicating on the two networks at a base station, such as when the SAS entity assigns to a base station a portion of the spectrum compatible for one network that is incompatible with the other network. Indeed, systems and methods that enable the network operators to indicate to the SAS entity when DSS operations are expected to be used are desired, so that the SAS entity is able to allocate portions of the radio spectrum (e.g., frequency ranges, channels) to accommodate respective network configurations.

(14) Enabling the network operator to notify the SAS entity of spectrum preferences and/or

expected operations may result in efficient deployment of communication networks since the SAS entity may assign a portion of the spectrum based on the spectrum preferences and/or the expected operation, thereby reducing a likelihood of the SAS entity unintentionally preventing the network operator from implementing DSS operations.

(15) In this way, when the network operator specifies to the SAS entity that the base station is to perform the first network (e.g., 4G/LTE) communications and not the second network (e.g., 5G/NR) communications, the SAS entity may allocate a portion of the spectrum aligned to a raster assigned to the first network communications (e.g., 100 kHz raster). When the operator specifies to the SAS entity that the base station is to perform the second network communications and not the first network communications, the SAS entity may allocate a portion of the spectrum aligned to a raster assigned to the first network communications (e.g., 30 kHz raster or 15 kHz raster). And, when the operator specifies to the SAS entity that the base station is to perform dynamic spectrum sharing to communicate using the first network (e.g., 4G/LTE) and the second network (e.g., 5G/NR) communications, the SAS entity may allocate a portion of the spectrum aligned to a raster assigned to both the first network communications and the second network communications (e.g., 300 kHz raster).

(16) Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:
- (2) FIG. 1 is a schematic block diagram of an electronic device including a transceiver, in accordance with an embodiment of the present disclosure;
- (3) FIG. 2 is a perspective view of a notebook computer representing a first embodiment of the electronic device of FIG. 1;
- (4) FIG. 3 is a front view of a handheld device representing a second embodiment of the electronic device of FIG. 1;
- (5) FIG. 4 is a front view of another handheld device representing a third embodiment of the electronic device of FIG. 1;
- (6) FIG. 5 is a front view of a desktop computer representing a fourth embodiment of the electronic device of FIG. 1;
- (7) FIG. 6 is a front view and side view of a wearable electronic device representing a fifth embodiment of the electronic device of FIG. 1;
- (8) FIG. 7 is an illustration of base stations communicating with an electronic device, such as the electronic device of FIG. 1, in accordance with an embodiment of the present disclosure;
- (9) FIG. 8 is an illustration of an example spectrum available for assignment to the base stations of FIG. 7, in accordance with an embodiment of the present disclosure;
- (10) FIG. 9A is an illustration of a first spectrum allocation of the example spectrum of FIG. 8, where each spectrum allocation request is granted, in accordance with an embodiment of the present disclosure;

(11) FIG. 9B is an illustration of a second spectrum allocation of the example spectrum of FIG. 8, where at least some spectrum allocation requests are not granted, in accordance with an embodiment of the present disclosure;

(12) FIG. 10 is a flowchart of a method for user equipment (e.g., the electronic device of FIG. 1) to communicate using an allocated portion (e.g., an assigned channel) of the spectrum of FIG. 8, in accordance with an embodiment of the present disclosure;

(13) FIG. 11 is a flowchart of a method for operating a base station to request and implement an allocated portion of the spectrum of FIG. 8, in accordance with an embodiment of the present disclosure;

(14) FIG. 12 is a flowchart of a method for operating a spectrum access system (SAS) entity to allocate a portion of the spectrum of FIG. 8 to the base station, in accordance with an embodiment of the present disclosure; and

(15) FIG. 13 is a flowchart of a method for operating a spectrum access system (SAS) entity to determine the portion of the spectrum of FIG. 8 to assign to the base station, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

(16) One or more specific embodiments of the present disclosure will be described below. These described embodiments are examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(17) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(18) Various processes are disclosed that may adjust an operating frequency range of an antenna. The processes may apply to a variety of electronic devices. It is noted that a channel may be a medium used to convey information from a sender (transmitter) to a receiver. It should be noted that since characteristics of the term “channel” may differ according to different wireless protocols, the term “channel” as used herein may be considered as being used in a manner that is consistent with the standard characterizing a device that the term “channel” is referencing. In some standards, channel widths may be variable (e.g., depending on device capability, band conditions). For example, long-term evolution (LTE) networks may support scalable channel bandwidths from 1.4 Megahertz (MHz) to 20 MHz. In contrast, wireless local area network (WLAN) channels may be 22 MHz wide while BLUETOOTH® channels may be 1 Mhz wide. Other protocols and standards may include different definitions of channels. Furthermore, some standards may define and use multiple types of channels, such as different channels for uplink or downlink and/or different channels for different uses such as data, control information, or the like. Also, as used herein, the term “band” has the full breadth of its ordinary meaning, and at least includes a section of spectrum (e.g., radio frequency spectrum) in which channels are used or set aside for the same purpose.

(19) Furthermore, in additional or alternative embodiments, a processor of an electronic device may indicate to a spectrum access system (SAS) entity what type of network that a network

operator is expecting to deploy a network used to communicate with one or more user equipment, such as a network enabled with spectrum sharing or a network not enabled with spectrum sharing. The SAS entity may assign the network operator a channel (or frequency range) compatible with two or more networks in response to receiving an indication that the network operator is expecting to deploy a network with spectrum sharing, such that the channel is able to be shared between two network types (e.g., shared between a 4G/LTE network and a 5G/NR network). With the foregoing in mind, a general description of suitable electronic devices that may include such a processor is provided below.

(20) Turning first to FIG. 1, an electronic device **10** according to an embodiment of the present disclosure may include, among other things, one or more of processor(s) **12**, memory **14**, nonvolatile storage **16**, a display **18**, input structures **22**, an input/output (I/O) interface **24**, a network interface **26**, a transceiver **28**, and a power source **30**. The various functional blocks shown in FIG. 1 may include hardware elements (including circuitry), software elements (including computer code stored on a computer-readable medium) or a combination of both hardware and software elements. Furthermore, a combination of elements may be included in tangible, non-transitory, and machine-readable medium that include machine-readable instructions. The instructions may be executed by the processor **12** and may cause the processor **12** to perform operations as described herein. It should be noted that FIG. 1 is merely one example of a particular embodiment and is intended to illustrate the types of elements that may be present in the electronic device **10**.

(21) By way of example, the electronic device **10** may represent a block diagram of the notebook computer depicted in FIG. 2, the handheld device depicted in FIG. 3, the handheld device depicted in FIG. 4, the desktop computer depicted in FIG. 5, the wearable electronic device depicted in FIG. 6, or similar devices. It should be noted that the processor **12** and other related items in FIG. 1 may be generally referred to herein as “data processing circuitry.” Such data processing circuitry may be embodied wholly or in part as software, firmware, hardware, or any combination thereof. Furthermore, the data processing circuitry may be a single contained processing module or may be incorporated wholly or partially within any of the other elements within the electronic device **10**.

(22) In the electronic device **10** of FIG. 1, the processor **12** may operably couple with the memory **14** and the nonvolatile storage **16** to perform various algorithms. Such programs or instructions executed by the processor **12** may be stored in any suitable article of manufacture that includes one or more tangible, computer-readable media at least collectively storing the instructions or processes, such as the memory **14** and the nonvolatile storage **16**. The memory **14** and the nonvolatile storage **16** may include any suitable articles of manufacture for storing data and executable instructions, such as random-access memory, read-only memory, rewritable flash memory, hard drives, and optical discs. Also, programs (e.g., an operating system) encoded on such a computer program product may also include instructions executable by the processor **12** to enable the electronic device **10** to provide various functionalities.

(23) In certain embodiments, the display **18** may be a liquid crystal display (LCD), which may facilitate users to view images generated on the electronic device **10**. In some embodiments, the display **18** may include a touch screen, which may facilitate user interaction with a user interface of the electronic device **10**. Furthermore, it should be appreciated that, in some embodiments, the display **18** may include one or more organic light emitting diode (OLED) displays, or some combination of LCD panels and OLED panels.

(24) The processors **12** (e.g., as part of or in the form of a controller) may operate circuitry to input or output data generated by the electronic device **10**. For example, the processors **12** may control and/or operate the memory **14**, the nonvolatile storage **16**, display **18**, input structures **22**, an input/output (I/O interface) **24**, a network interface **26**, a transceiver **28**, a power source **29**, or the like to perform operations of the electronic device **10** and/or to facilitate control of the operations of the electronic device. In particular, the processors **12** may generate control signals for operating

the transceiver **28** to transmit data on one or more communication networks.

(25) The input structures **22** of the electronic device **10** may enable a user to interact with the electronic device **10** (e.g., pressing a button to increase or decrease a volume level). The I/O interface **24** may enable the electronic device **10** to interface with various other electronic devices, as may the network interface **26**. The network interface **26** may include, for example, one or more interfaces for a personal area network (PAN), such as a BLUETOOTH® network, for a local area network (LAN) or wireless local area network (WLAN), such as an 802.11x WI-FI® network, and/or for a wide area network (WAN), such as a 3.sup.rd generation (3G) cellular network, 4.sup.th generation (4G) cellular network, LTE cellular network, long term evolution license assisted access (LTE-LAA) cellular network, 5.sup.th generation (5G) cellular network, or New Radio (NR) cellular network. The network interface **26** may also include one or more interfaces for, for example, broadband fixed wireless access networks (e.g., WIMAX®), mobile broadband Wireless networks (mobile WIMAX®), asynchronous digital subscriber lines (e.g., ADSL, VDSL), digital video broadcasting-terrestrial (DVB-T®) network and its extension DVB Handheld (DVB-H®) network, ultra-wideband (UWB) network, alternating current (AC) power lines, and so forth.

(26) In some embodiments, the electronic device **10** communicates over the aforementioned wireless networks (e.g., WI-FI®, WIMAX®, mobile WIMAX®, 4G, LTE®, 5G, and so forth) using the transceiver **28**. The transceiver **28** may include circuitry useful in both wirelessly receiving and wirelessly transmitting signals (e.g., data signals, wireless data signals, wireless carrier signals, RF signals), such as a transmitter and/or a receiver. Indeed, in some embodiments, the transceiver **28** may include a transmitter and a receiver combined into a single unit, or, in other embodiments, the transceiver **28** may include a transmitter separate from a receiver. The transceiver **28** may transmit and receive RF signals to support voice and/or data communication in wireless applications such as, for example, PAN networks (e.g., BLUETOOTH®), WLAN networks (e.g., 802.11x WI-FI®), WAN networks (e.g., 3G, 4G, 5G, NR, and LTE® and LTE-LAA cellular networks), WIMAX® networks, mobile WIMAX® networks, ADSL and VDSL networks, DVB-T® and DVB-H® networks, UWB networks, and so forth. As further illustrated, the electronic device **10** may include the power source **30**. The power source **30** may include any suitable source of power, such as a rechargeable lithium polymer (Li-poly) battery and/or an alternating current (AC) power converter.

(27) In certain embodiments, the electronic device **10** may take the form of a computer, a portable electronic device, a wearable electronic device, or other type of electronic device. Such computers may be generally portable (such as laptop, notebook, and tablet computers) and/or those that are generally used in one place (such as desktop computers, workstations and/or servers). In certain embodiments, the electronic device **10** in the form of a computer may be a model of a MACBOOK®, MACBOOK® PRO, MACBOOK AIR®, IMAC®, MAC® mini, or MAC PRO® available from Apple Inc. of Cupertino, California. By way of example, the electronic device **10**, taking the form of a notebook computer **10A**, is illustrated in FIG. 2 in accordance with one embodiment of the present disclosure. The notebook computer **10A** may include a housing or the enclosure **36**, the display **18**, the input structures **22**, and ports associated with the I/O interface **24**. In one embodiment, the input structures **22** (such as a keyboard and/or touchpad) may enable interaction with the notebook computer **10A**, such as starting, controlling, or operating a graphical user interface (GUI) and/or applications running on the notebook computer **10A**. For example, a keyboard and/or touchpad may facilitate user interaction with a user interface, GUI, and/or application interface displayed on display **18**.

(28) FIG. 3 depicts a front view of a handheld device **10B**, which represents one embodiment of the electronic device **10**. The handheld device **10B** may represent, for example, a portable phone, a media player, a personal data organizer, a handheld game platform, or any combination of such devices. By way of example, the handheld device **10B** may be a model of an IPOD® or IPHONE® available from Apple Inc. of Cupertino, California. The handheld device **10B** may include the

enclosure **36** to protect interior elements from physical damage and to shield them from electromagnetic interference. The enclosure **36** may surround the display **18**. The I/O interface **24** may open through the enclosure **36** and may include, for example, an I/O port for a hard wired connection for charging and/or content manipulation using a connector and protocol, such as the Lightning connector provided by Apple Inc. of Cupertino, California, a universal serial bus (USB), or other similar connector and protocol.

(29) The input structures **22**, in combination with the display **18**, may enable user control of the handheld device **10B**. For example, the input structures **22** may activate or deactivate the handheld device **10B**, navigate a user interface to a home screen, present a user-editable application screen, and/or activate a voice-recognition feature of the handheld device **10B**. Other of the input structures **22** may provide volume control, or may toggle between vibrate and ring modes. The input structures **22** may also include a microphone to obtain a user's voice for various voice-related features, and a speaker to enable audio playback. The input structures **22** may also include a headphone input to enable input from external speakers and/or headphones.

(30) FIG. **4** depicts a front view of another handheld device **10C**, which represents another embodiment of the electronic device **10**. The handheld device **10C** may represent, for example, a tablet computer, or one of various portable computing devices. By way of example, the handheld device **10C** may be a tablet-sized embodiment of the electronic device **10**, which may be, for example, a model of an IPAD® available from Apple Inc. of Cupertino, California.

(31) Turning to FIG. **5**, a computer **10D** may represent another embodiment of the electronic device **10** of FIG. **1**. The computer **10D** may be any computer, such as a desktop computer, a server, or a notebook computer, and/or may be a standalone media player or video gaming machine. By way of example, the computer **10D** may be an IMAC®, a MACBOOK®, or other similar device by Apple Inc. of Cupertino, California. It should be noted that the computer **10D** may also represent a personal computer (PC) by another manufacturer. The enclosure **36** may protect and enclose internal elements of the computer **10D**, such as the display **18**. In certain embodiments, a user of the computer **10D** may interact with the computer **10D** using various peripheral input devices, such as keyboard **22A** or mouse **22B** (e.g., input structures **22**), which may operatively couple to the computer **10D**.

(32) Similarly, FIG. **6** depicts a wearable electronic device **10E** representing another embodiment of the electronic device **10** of FIG. **1**. By way of example, the wearable electronic device **10E**, which may include a wristband **43**, may be an APPLE WATCH® by Apple Inc. of Cupertino, California. However, in other embodiments, the wearable electronic device **10E** may include any wearable electronic device such as, a wearable exercise monitoring device (e.g., pedometer, accelerometer, heart rate monitor), or other device by another manufacturer. The display **18** of the wearable electronic device **10E** may include a touch screen version of the display **18** (e.g., LCD, OLED display, active-matrix organic light emitting diode (AMOLED) display, and so forth), as well as the input structures **22**, which may facilitate user interaction with a user interface of the wearable electronic device **10E**. In certain embodiments, as previously noted above, each embodiment (e.g., notebook computer **10A**, handheld device **10B**, handheld device **10C**, computer **10D**, and wearable electronic device **10E**) of the electronic device **10** may include the transceiver **28**.

(33) Keeping the foregoing in mind, FIG. **7** is an illustration of access network nodes communicating with the electronic device **10**, according to embodiments of the present disclosure. The access network nodes may include devices such as an eNodeB (eNB) for long-term evolution (LTE) access networks and/or a next generation NodeB (gNB) for 5th generation (5G) access networks. For ease of discussion, eNBs and gNBs, and any other network nodes for various other communication networks, are referred to as base stations **50** (e.g., base station **50A**, base station **50B**). FIG. **7** also depicts user equipment (UE), such as an electronic device **52** (which may include the electronic device **10**). Each of the base stations **50** and/or the electronic device **52** may have one

or more components similar to the electronic device **10**, and thus may include control circuitry, such as the processor **12**, and/or memory circuitry, such as the memory **14** and/or nonvolatile storage **16**, which may operate together to respectively cause the base stations **50** and/or the electronic device **52** to perform respective operations.

(34) It is noted that user equipment able to communicate with the base stations **50** may include any of various types of computer systems device which are mobile or portable and which performs wireless communications. Examples of user equipment any suitable portable electronic devices, mobile telephones, smart phones, portable gaming devices, laptops, wearable devices, or the like. In general, the term “user equipment,” “UE”, or “UE device” may broadly be defined to encompass any electronic, computing, and/or telecommunications device (or combination of devices) transportable by a user and capable of wireless communication, including the electronic device **52**.

(35) A network operator **54** (used interchangeably with “network provider”) may use the base stations **50** as physical communication nodes to deploy a radio network managed by the network operator **54** over one or more cells (e.g., regions) managed by each respective base station **50**. In particular, each of the base stations **50** may be associated with one or more cells or regions and thus may provide access to the radio network for each electronic device **52** physically or logically located in the cell or region that the respective base station **50** serves. The term “base station” has the full breadth of its ordinary meaning, and at least includes a wireless communication station installed at a location that is used to communicate as part of a wireless telephone system or radio system, such as the radio network managed by the network operator.

(36) The base stations **50** and the electronic device **52** may communicate over a transmission medium (e.g., air, atmosphere, water) using any of various radio access technologies (RATs), also referred to as wireless communication technologies or telecommunication standards, including Global System for Mobile Communications (GSM), Universal Mobile Telecommunications Service (UMTS) associated with, for example, Wideband Code Division Multiple Access (WCDMA) or Time Division Synchronous Code Division Multiple Access (TD-SCDMA) air interfaces, Long-Term Evolution (LTE), LTE-Advanced (LTE-A), 5G New Radio (5G NR), High Speed Packet Access (HSPA), 3rd Generation Partnership Project (3GPP), 3.sup.rd Generation Partnership Project 2 (3GPP2), Code Division Multiple Access (CDMA), Code Division Multiple Access 2000 (CDMA2000) (e.g., Single-Carrier Radio Transmission Technology (1×RTT) air interface standard, Evolution-Data Optimized (1×EV-DO), High-Rate Packet Data (HRPD), Evolved High Rate Packet Data (eHRPD)), or the like. Note that if a respective base station of the base stations **50** is implemented in the context of LTE, it may alternately be referred to as an “eNodeB” or “eNB”, and that if a respective base station of the base stations is implemented in the context of 5G NR, it may alternately be referred to as “gNodeB” or “gNB.”

(37) Thus, base stations **50** may act as “serving cells” for electronic devices **52** disposed or located (e.g., logically, physically) within a cell, where the cell may be defined as where the base station **50** is located or nearby (e.g., a threshold distance from the base station **50**). Indeed, each serving cell may communicate with user equipment using a subset of a radio spectrum dedicated to that communicative coupling, sometimes referred to as a “subcarrier”. A subcarrier may be dedicated to communications between the electronic device **52** and a network operator **54** via the base stations **50**, such as base station **50B**. In some cases, the subcarrier may be referred to interchangeably as a “component carrier,” a “sub-carrier,” and/or a “sub-channel” and may be managed by the base station **50B**. Thus, the base station **50B**, as a serving cell, serves as the radio access point on behalf of a network operator **54** to enable communication to occur between the network operator **54** and an electronic device **52** while the electronic device **52** is located within boundaries (e.g., logically-defined boundaries, volume-defined boundaries, area-defined boundaries) of a cell managed by the network operator **54** via the corresponding base station **50**.

(38) The base stations **50**, when deploying one or more networks on behalf of network operators, may respectively communicate with user equipment (e.g., electronic devices **52**) using a first

network (e.g., 4G/LTE), a second network (e.g., 5G/NR), or both. For example, the base station **50B** may communicate with the electronic device **52A** using the first network (e.g., 4G/LTE), and may communicate with a second electronic device **52B** (e.g., additional user equipment) using the second network (e.g., 5G/NR) when capable of performing DSS operations and/or when allocated two portions of a radio spectrum (e.g., one portion to be used for the first network and the other portion to be used for the second network). Since a limited number of frequencies exist within a frequency spectrum, networks may be more efficiently deployed when a spectrum access system (SAS) entity **56** enables communication using both the first network and the second network (or any number of shared networks) by assigning a frequency range compatible with both communications on the first network and on the second network. When using a frequency compatible with both, for example when the frequency range aligns with a spectrum raster for both the first network and the second network, a respective base station of the base stations **50** may use both networks without resetting the channel (e.g., adjusting frequency ranges) used for communicating when changing the network used for communication.

(39) It is noted that these systems and methods may be used to deployed networks of different types than 4G/LTE or 5G/NR. For example, these systems and methods may be used to provide a spectrum shared between a 2G network, a 3G network, and a 4G network, or any other suitable combination of networks. It is also noted that several variations of communication interconnections between the network operators **54**, the base stations **50**, SAS entity **56**, and the electronic devices **52** is shown in FIG. 7. In this way, any suitable communicative couplings may be used to interconnect the components of the depicted system to facilitate operation and/or communication between the components. For example, in one network system, the network operator **54** may communicate directly with the SAS entity **56**, however in another network system, the network operator **54** may sometimes communicate with the SAS entity **56** or exclusive communicate with the SAS entity **56** via the base station. Other combinations of these communicative couplings not depicted may also be compatible with the descriptions herein.

(40) To elaborate, FIG. 8 is an illustration of an example radio spectrum **70** available for assignment to the base stations **50**, according to embodiments of the present disclosure. When communicating with the electronic device **52**, a respective base station, for example the base station **50B**, may transmit messages on subsets or “sub-spectrums” of a total radio spectrum (e.g., spectrum, frequency spectrum). In some embodiments, a few portions **72** (e.g., portion **72A**, portion **72B**) of the total radio spectrum **70** may be reserved, such as for federal use (e.g., for use by a government), while another portion **74** of the total radio spectrum **70** may be shared between multiple network operators, such as incumbent access operators, priority access operators, and/or general authorized access operators. Indeed, the portion **74** may be dedicated as a general-use and/or an open-access band, such as a Citizens Broadband Radio Service (CBRS) band. The portion **74** may be divided into any number of channels **76**, where each channel may be respectively assigned to one or more network operators. The various network operators may, in some cases, each correspond to an assigned priority used by an SAS entity **56** when prioritizing allocations of the spectrum portion **74**. For example, in determining a sequence or order for which each network operator is allocated one or more channels **76**, the SAS entity **56** may determine that incumbent access operators are granted higher priority than the priority access operators, and that the priority access operators are granted higher priority than the general authorized access operators.

(41) The channels **76** may be defined by the SAS entity **56** via channel parameters **78**, including center frequency **78A** and a bandwidth **78B**. The bandwidth **78B** may be defined by a frequency band, which may include one or more of the frequency ranges and be delimited by a lower frequency and a higher frequency. Channel rasters **80** may respectively define a channel center frequency **78A** for a channel. In this way, different networks employing different channel width configurations may use various of the channels **76** due to the channels **76** having different

frequency increments between the channel raster **80**. That is, the SAS entity **56** may dynamically set a channel raster **80** as a channel center frequency **78A** when dynamically allocating a respective channel **76** to a network operator.

(42) Furthermore, each channel **76** may be further sub-divided into sub-channels (not pictured), where each sub-channel may have a respective center frequency and respective bandwidth. Messages transmitted using frequencies within a same frequency band (e.g., channel **76**) but using separate sub-channels (e.g., component carriers) having different frequency ranges within the same frequency band may be transmitted (e.g., concurrently) without cross-interference.

(43) Some frequency ranges may be defined by a standards body (e.g., standards generated by the Third Generation Partnership Project (3GPP) standards body or development group) as for a specific use (e.g., purpose) and/or for a specific type of communication. That is, standards bodies may define frequency bands for different communication protocols and/or communication networks, and thus may include different frequency bands for a 3rd generation (3G) frequency band, 4th generation (4G) frequency band, 5th generation (5G) frequency band, or the like. For example, frequencies between 24 Gigahertz (GHz) and 48 GHz may be reserved for 5G/NR network communications. However, as illustrated, a narrow band (e.g., frequencies between 3.5 GHz and 3.6 GHz, frequencies between 3550 MHz and 3650 MHz) may be reserved as the general-use band (e.g., portion **74**), such as the CBRS band corresponding to the spectrum portion **74** described-above.

(44) Indeed, a first network operator that deploys a first network (e.g., a 4G/LTE network) and a second network operator that deploys a second network (e.g., a 5G/NR network) may both use the general-use band for communication. The SAS entity **56** may arbitrate access to portions of the radio spectrum portion **74** between the various network operators requesting access to a respective portion of the radio spectrum portion **74** (e.g., based on the respective network). Each network operator may submit a preference to the SAS entity **56** when requesting a frequency band for communication via a spectrum grant request. The SAS entity **56** may consider preferences when assigning channels **76**, but may additionally or alternatively consider relative priorities of the network operators requesting access.

(45) Thus, the presently disclosed techniques may enable network operators to notify the SAS entity **56** of their respective communication capabilities and preferences when requesting a channel **76** for their network. This may include, for example, the network operators enhancing a grant request message transmitted to the SAS entity **56** when requesting an allocation of the spectrum portion **74**. The network operators may enhance the grant request message with information that indicates to the SAS entity **56** a network type used by the network operator (e.g., DSS deployment, non-DSS deployment).

(46) To request a channel **76**, the network operator may, using one of the base stations **50**, such as base station **50B**, transmit a registration request to the SAS entity **56** and receive a response from the SAS entity **56** confirming the registration. The network operator may, using the base station **50B**, continue by transmitting a spectrum grant request that includes a usage type parameter. The network operator may, using the base station **50B**, may additionally or alternatively transmit additional information regarding communication capabilities and preferences for reference by the SAS entity **56** when allocating a channel to the network operator. The usage type parameter may indicate what networks the network operator is expecting to use (e.g., a first network, a second network, or both in DSS operations). The SAS entity **56** may respond to the network operator, via the base station **50B**, to confirm receipt of the spectrum grant request. The SAS entity **56** may transmit the spectrum allocation to the network operator, via the base station **50B**, after determining which channel **76** to allocate to the network operator based on the usage type parameter and/or the additional information included in the spectrum grant request. For example, the spectrum allocation sent by the SAS entity **56** may include an indication of a channel **76** defined as a frequency range to be used for communication.

(47) To elaborate, FIG. 9A is an illustration of a first spectrum allocation example operation of the SAS entity 56 determining to allocate respective channels 76 of the spectrum 70 shown in FIG. 8 to multiple network operators 92, according to embodiments of the present disclosure. A first operator 92A (e.g., a respective network operator of the network operators 92) may correspond to a highest priority of the SAS entity 56, a second operator 92B, a third operator 92C, and a fourth operator 92D may each correspond to a middle priority of the SAS entity 56, and the remaining operators (e.g., fifth operator 92E, sixth operator 92F, seventh operator 92G) may correspond to a lowest priority of the SAS entity 56. Each operator 92 may request a spectrum allocation from the SAS entity 56 based on which communication operations the respective operator network desires to perform and a desired bandwidth. In this example, the operators 92 request spectrum allocation based on LTE-exclusive communication operations, NR-exclusive communication operations, or LTE/NR DSS communication operations (where both LTE and NR may be used to communicate) in their respective spectrum grant requests. The SAS entity 56, in response to the spectrum grant request, determines a channel compatible with parameters of the spectrum grant request, such as compatible with a raster of the network to be used by the respective operator 92, and assigns the channel to the respective operator 92.

(48) A raster defines a reference radio frequency in a frequency range that may be used to identify the uplink and downlink channel positions of a given network type for a particular network operator. Each channel center frequency may be a multiple of a raster. Rasters may include frequency intervals that UEs use to scan channels 76 to synchronize with a network, to identify themselves with the network, acquire knowledge regarding a network that the UE is attempting to connect with, or the like.

(49) In this way, each channel available for assignment to a particular network operator corresponds to a multiple of a raster used to provide a network type of the network to be deployed by the network operator. For example, when the network operator is to deploy a 4G/LTE network, the raster used is 100 kHz, while when the network operator is to deploy a 5G/NR network, the raster used is either 15 kHz or 30 kHz based on, for example, particulars of the network deployment and/or operating policies of the SAS entity 56 defining which raster to use to assign channels. Since different network types may communicate using relatively more narrowband or more wideband communications, different networks may use different rasters to define eligible channels for allocation to a respective network operator.

(50) To elaborate, the first operator 92A may request, via a spectrum grant request, a channel 76 compatible with DSS operations for a first network and a second network (e.g., compatible with both LTE communications and NR communications), and a bandwidth allocation of at least 20 MHz from the SAS entity 56. Based on the request for a DSS operation compatible channel, the SAS entity 56 may identify which of the channels 76 are characterized by a center frequency that aligns with a raster for the first network and a raster for the second network. For example, the SAS entity 56 assigns to the first operator 92A a 20 MHz frequency range referred to as a channel 94 (e.g., a channel defined between approximately 3590 MHz and 3610 MHz having a center frequency at 3600 MHz) that was determined to be compatible with DSS operations. In particular, the channel 94 may be compatible with DSS operations for the 4G/LTE network and the 5G/NR network because the center frequency of 3600 MHz (i.e., 3,600,000 kHz) aligns with a 300 kHz raster, which is a multiple of both the 100 kHz raster of the 4G/LTE network and the 15 kHz or 30 kHz of the 5G/NR network. That is, the channel 94 is characterized (e.g., includes) a center frequency aligned on a raster that is a multiple of respective rasters of the first network and the second network. Comparing FIG. 9A to FIG. 8, the channel 94 extends beyond the 10 MHz channels (e.g., channels 76) shown in FIG. 8. The SAS entity 56 may thus allocate any portion of the spectrum portion 74 to any operator that the SAS entity 56 deems suitable.

(51) The second operator 92B and the third operator 92C may request, via respective spectrum grant requests to the SAS entity 56, respective channels 76 compatible with DSS operations (e.g.,

compatible with both LTE communications and NR communications), and bandwidth allocations of at least 10 MHz. The SAS entity **56** may assign a channel **96** to the second operator **92B**, and assign a channel **98** to the third operator **92C**, each channel (e.g., channel **96**, channel **98**) being compatible with DSS operations (e.g., raster aligning on 300 kHz) and having a bandwidth allocation substantially equal to 10 MHz. In particular, the channel **96** is assigned a range of frequencies between approximately 3550 MHz and 3560 MHz having a center frequency at 3555 MHz, and the channel **98** is assigned a range of frequencies between approximately 3610 MHz and 3620 MHz having a center frequency at 3615 MHz. In particular, the channels **96**, **98** may be compatible with DSS operations for the 4G/LTE network and the 5G/NR network because the center frequencies of 3555 MHz (i.e., 3,555,000 kHz) and 3615 MHz (i.e., 3,615,000 kHz) align with a 300 kHz raster, which is a multiple of both the 100 kHz raster of the 4G/LTE network and the 15 kHz or 30 kHz of the 5G/NR network.

(52) The fourth operator **92D** may request, via a spectrum grant request to the SAS entity **56**, or a channel **76** compatible with LTE operations, and a bandwidth allocation of at least 20 MHz. The SAS entity **56** may assign a channel **100** to the fourth operator **92D**, where the channel **100** is assigned a range of frequencies equal to 20 MHz between approximately 3570 MHz and 3590 MHz having a center frequency at 3580 MHz. The center frequency for the channel **100** is selected by the SAS entity **56** for the fourth operator **92D** since the center frequency of 3580 MHz (i.e., 3,580,000 kHz) aligns with one of many eligible center frequencies for use with LTE communications (e.g., raster aligning on 100 kHz designated for LTE communications) without aligning for one of many eligible center frequencies for use with NR communications, and thus cannot be used for NR communications. It is noted that the 20 MHz allocations (e.g., channel **100**, channel **94**) are depicted as approximately twice the width as the 10 MHz allocations (e.g., channel **96**) in FIG. **9A**. The SAS entity **56** may assign to the remaining operators (e.g., fifth operator **92E**, sixth operator **92F**, seventh operator **92G**) 10 MHz allocations at center frequencies of 3625 MHz (e.g., channel **102**), 3635 MHz (e.g., channel **104**), and 3645 MHz (e.g., channel **106**) in response to preferences submitted by these operators via respective spectrum grant requests. In particular, the preferences submitted by these operators may have indicated that the fifth operator **92E**, the sixth operator **92F**, and the seventh operator **92G** each desired a 10 MHz channel able to be used for non-DSS, LTE-exclusive network deployments (e.g., communications). The channels, such as the channel **102**, the channel **104**, and the channel **106**, may be considered unallocated frequency ranges before being assigned to one of the network operators. The channel **102** is assigned a range of frequencies between approximately 3620 MHz and 3630 MHz having a center frequency at 3625 MHz (i.e., 3,625,000 kHz) and the channel **104** is assigned a range of frequencies between approximately 3630 MHz and 3640 MHz having a center frequency at 3635 MHz (i.e., 3,635,000 kHz). Furthermore, the channel **106** is assigned a range of frequencies between approximately 3640 MHz and 3650 MHz having a center frequency at 3645 MHz (i.e., 3,645,000 kHz). It is noted that, in some cases, one network operator may transmit multiple requests, such as one request from multiple base stations **50** and/or multiple requests from a single base station **50**. In these cases, the SAS entity **56** may assign more than one channel **76** to the network operator to assign relatively larger amounts of frequency ranges to the network operator.

(53) In some cases, however, the SAS entity **56** may not be able to grant each request and/or indicated preference of each network operator. For example, FIG. **9B** is an illustration of a second spectrum allocation example, according to embodiments of the present disclosure. Presuming same priorities as in FIG. **9A** and same requests for each operator **92** discussed, in some cases, the SAS entity **56** changes an allocation and does not grant the exact allocation requested.

(54) For example, the first operator **92A** may request, via a spectrum grant request, a channel **76** compatible with DSS operations (e.g., compatible with both LTE communications and NR communications) and a bandwidth allocation of at least 20 MHz. Instead, the SAS entity **56** may partially grant the request of the first operator **92A**. This may happen when, for example, the SAS

entity **56** determines that a portion of the spectrum portion **74** (e.g., channel **108**) has already been allocated to a network operator (e.g., eighth operator **92H**), where the portion (e.g., channel **108**) may have otherwise matched a configuration indicated in the spectrum grant request from the first operator **92A**. Thus, the SAS entity **56** may grant the first operator **92A** a 10 MHz channel having a center frequency of 3645 MHz as a channel **110** as opposed to the channel **94** (e.g., a 20 MHz channel having a center frequency of 3600 MHz exemplified in FIG. **9A**). The SAS entity **56** may determine that the center frequency of 3645 MHz is compatible with DSS operations since the center frequency of 3645 MHz is aligned to a raster of both LTE communications and NR communications (e.g., 300 kHz raster used for DSS operations of LTE and NR communications). To make room for the channel **110** using a relatively higher frequency of the spectrum portion **74** than that of the channel **94** in FIG. **9A** for the first operator **92A**, the SAS entity **56** may allocate channel assignments for the other operators **92** similarly as described in FIG. **9A**, except the channel allocations for the fifth operator **92E**, sixth operator **92F**, and seventh operator **92G**. These channel allocations may continue to be 10 MHz allocations compatible with LTE communications and incompatible with NR communications (due to the raster alignments of the center frequencies). However, the SAS entity **56** may shift the channel allocations to have center frequencies of 3625 MHz (e.g., channel **104**), 3635 MHz (e.g., channel **106**), and 3565 MHz (e.g., channel **102**), respectively. It is noted that the SAS entity **56** moving the channel **102** to a lower frequency than the channel **104** or the channel **106** is merely an example, and the SAS entity **56** may instead perform any order of shifting or no shifting at all.

(55) As described above, the base stations **50**, as a proxy and/or representative actor of the network operator **54** (e.g., network provider), may transmit communication preferences to the SAS entity **56** when registering to (e.g., requesting) a spectrum allocation of the spectrum **70** for use with communication with user equipment requesting access to a network deployed by the network operator **54**. In this way, methods described in reference to FIG. **10-13** elaborate on communications and/or operators of the user equipment, base station, and SAS entity **56** to provide further explanation of these systems. FIG. **10** is a flowchart of a method **120** for operating user equipment (e.g., electronic device **10**, electronic device **52**) to communicate using an allocated portion of the spectrum **70** (e.g., an assigned channel), according to embodiments of the present disclosure. It is noted that, although depicted in a particular order, the blocks of the method **120** may be performed in any suitable order, and at least some blocks may be skipped altogether. As described herein, the method **120** is described as performed by the electronic device **52**, however, it should be understood that any suitable processing and/or control circuitry may perform some or all of the operations of the method **120**, such as one or more of the processors **12**.

(56) At block **122**, the electronic device **52** detects one of the base stations **50**. For ease of discussion, the method **120** is discussed in terms of electronic device **52** as an exemplary user equipment and base station **50B** as an exemplary radio access node (e.g., example base station). The electronic device **52** may detect the base stations **50B** many different ways, including for example, by receiving one or signals emitted from the base station **50B**. As discussed above, the base stations **50** shown in FIG. **7** may each correspond to one or more network operators **54**. One network operator **54** may deploy its communication network via each of the depicted base stations **50**, or multiple network operators **54** may deploy multiple networks via respective of the base stations **50**. Indeed, two or more base stations **50** may correspond to one network operator **54**, and communicate with user equipment while the user equipment is disposed or located within its serving cells as discussed above.

(57) At block **124**, the electronic device **52** transmits a registration request to the base station **50B**. The electronic device **52** may transmit the registration request in response to detecting the base station **50B**. The registration request may be associated with operations to register or authenticate a subscriber identification module (SIM) card of the electronic device **52** to the base station **50B** and/or to a network operator **54** deploying a radio network using the base station **50B**. The base

station **50B** may use information of the registration request, such as identifiers and/or authentication keys, to verify that the electronic device **52** is permitted to communicate using the radio network.

(58) At block **126**, the electronic device **52** receives a communication configuration from the base station **50B** that includes a spectrum allocation assignment. The spectrum allocation assignment may correspond to an assignment determined by the SAS entity **56**, and thus may correspond to one of many channels **76** described with reference to FIGS. **8-9B**. In some cases, the spectrum allocation assignment indicates the center frequency corresponding to the channel **76** assigned for communication. In other cases, however, the electronic device **52** may receive an indication of a subcarrier of the channel **76** particularly allocated by the base station **50B** for use in communications with the electronic device **52**. The indication of the subcarrier may correspond to a frequency range within a respective channel **76** to be used for communication between the electronic device **52** and the base station **50B** (where other electronic devices communicating with the base station **50B** may use a different subcarrier of the channel **76** for transmitting of messages). The communication configuration may also include uplink allocation timing, downlink allocation timing, other definition of communication slots and/or timings, or the like, for the electronic device **52** to reference when scheduling transmit operations or receive operations.

(59) At block **128**, the electronic device **52** communicates with the base station **50B** using the spectrum allocation assignment. Indeed, the electronic device **52** may communicate at least partially concurrently or simultaneously with one or more other electronic devices **10** communicating with the base station **50B**. The electronic device **52** may align messages for uplink operations to a center frequency associated with the center frequency of the subcarrier and/or of the channel **76** indicated by the spectrum allocation assignment. Furthermore, when receiving downlink communications, the electronic device **52** may anticipate messages being downloaded using the center frequency.

(60) To elaborate on complementary operations to the method **120** with respect to a base station **50**, FIG. **11** is a flowchart of a method **140** for operating a respective base station of the base stations **50**, such as the base station **50B**, to request and use an allocated portion of the radio spectrum **70**, according to embodiments of the present disclosure. It is noted that, although depicted in a particular order, the blocks of the method **140** may be performed in any suitable order, and at least some blocks may be skipped altogether. As described herein, the method **140** is described as performed by the base station **50B**, however, it should be understood that any suitable processing and/or control circuitry may perform some or all of the operations of the method **140**, such as the electronic device **10** which may include the base station **50**. As such, operations of the method **140** may be performed by the one or more processors **12**.

(61) At block **142**, the base station **50B** receives a registration request from the electronic device **52** and transmits an indication of the registration request (or a modified version of the registration request) to the network operator **54** to verify that the electronic device **52** is eligible for communication with the base station **50B**. The indication may include at least some of the information passed along in the registration request, including authentication information stored on a SIM card (e.g., a SIM card identifier used to identify a device using the SIM card as authorized user equipment) of the electronic device **52** for use by the network operator **54** and/or base station **50B** when verifying the electronic device **52**. For example, the network operator **54** may verify that the electronic device **52** is authorized to communicate on its network. Indeed, when the electronic device **52** enters a boundary of a corresponding cell, the base station **50B** may receive the registration request asking to initiate communication with the electronic device **52**. The electronic device **52** may transmit the registration request to the base station **50B** in response to determining that is the electronic device **52** itself is within a boundary of one of the cells managed (e.g., served) by the base station **50B**, in response to a request for registration received from the base station **50B**, or the like. Indeed, the base station **50B**, in some cases, may transmit the request for registration to

the electronic device **52** to indicate that the user equipment has entered a serving cell managed by the network operator **54**.

(62) At block **144**, the base station **50B** receives confirmation of registration from the network operator **54**. That is, the network operator **54** may confirm that the electronic device **52** is registered to the network managed by the network operator **54**. In some cases, this operation may be skipped, such as when the base station **50B** communicates with other base stations **50** regarding a registration status of the electronic device **52** to the network (e.g., such that each respective authentication operation is bypassed after the electronic device **52** is verified once).

(63) At block **146**, the base station **50B** transmits a spectrum grant request to an SAS entity **56** and may receive a channel allocation assignment from the SAS entity **56** in response to its request. The spectrum grant request may include a usage type parameter indicative of an operation to be implemented by the network operator **54** via the base station **50B**, and more particularly indicative of whether the network operator **54** desires to perform DSS or non-DSS operations and indicative of what networks are to be used by the network operator **54**. In some embodiments, the transmission of the spectrum grant request may occur prior to at least some operations of the block **142**, such as in response to the base station **50B** initially powering-on and/or detecting a first electronic device **52**. Once determined, a channel allocation assignment may be used for multiple communication operations between multiple electronic devices **52**, such as until the channel allocation assignment expires and/or a message is received from the SAS entity **56** indicating to the base station **50B** to pause (or stop) use of the channel allocation assignment. This may happen, for example, in response to a higher priority network operator **54** requesting a same channel allocation assignment as the allocated channel assignment. The channel assignment may be allocated to meet any suitable raster requirements to fulfill the operation indicated via the usage type parameter.

(64) At block **148**, the base station **50B** determines the spectrum allocation assignment for the electronic device **52** from the channel allocation assignment. In this way, the base station **50B** may assign a sub-channel from the bandwidth of the channel **76** indicated by the channel allocation assignment as assigned by the SAS entity **56** to the network operator **54**. The base station **50B** may determine the sub-channel based on which next sub-channel is to be used, based on a priority of the electronic device **52** relative to other devices of the network, or the like. In some cases, the base station **50B** may assign the whole channel **76** to the user equipment, or may assign two or more sub-channels of the channel **76** to the user equipment. Furthermore, in some cases, the base station **50B** may vary a width of sub-channels assigned to the user equipment. For example, a first user equipment may be assigned a channel having a 3 MHz bandwidth and a second user equipment may be assigned a channel having a 5 MHz bandwidth.

(65) At block **150**, the base station **50B** transmits a communication configuration to the electronic device **52** that includes the spectrum allocation assignment for the electronic device **52**. The communication configuration may include other information, such as indications of downlink allocations, uplink allocations, a duration of time for which the sub-channel assignment is valid, and the like. For example, the base station **50B** (at a later time during communication with the electronic device **52**) may access the communication configuration corresponding to the electronic device **52** to determine a message as incoming from the first user equipment (i.e., to determine an uplink allocation is upcoming). In response to determining the message as incoming, the base station **50B** may pause transmission of another message to the electronic device **52** (e.g., may pause downlink communications as to not interrupt or interfere with the uplink communication from the electronic device **52**). The base station **50B** may then receive the message from the electronic device **52**, such as via a sub-channel assigned to the electronic device **52** in the communication configuration (e.g., spectrum allocation assignment). The base station **50B** may also access the communication configuration to determine when a next downlink allocation is scheduled (e.g., to determine an upcoming downlink allocation to start at a first time), and may transmit the message that was paused at the start of the next downlink allocation (e.g., may transmit the paused message

at the first time) to the electronic device **52**.

(66) To elaborate on complementary operations to the method **120** and to the method **140** with respect to the SAS entity **56**, FIG. **12** is a flowchart of a method **164** for operating the SAS entity **56** to allocate a portion of the spectrum portion **74** to the base station **50B**, according to embodiments of the present disclosure. It is noted that, although depicted in a particular order, the blocks of the method **164** may be performed in any suitable order, and at least some blocks may be skipped altogether. As described herein, the method **164** is described as performed by the SAS entity **56**, however, it should be understood that any suitable processing and/or control circuitry may perform some or all of the operations of the method **164**, such as an electronic device **10** acting as the SAS entity **56** and using one or more of the processors **12** to perform operations of the method **164**.

(67) At block **166**, the SAS entity **56** receives the spectrum grant request from the base station **50B** (e.g., transmitted at block **146** of FIG. **11**). The spectrum grant request may include information such as whether the base station **50B** is expecting to perform DSS operations or non-DSS operations, the one or more networks that the base station **50B** is expecting to communicate on (e.g., 5G/NR and/or 4G/LTE), a preferred bandwidth for the channel allocation (e.g., 10 MHz, 20 MHz), and/or the like. Indeed, the spectrum grant request may include the usage type parameter that indicates an operation to be implemented by the network operator **54**, such as whether the operations are to be DSS operations or non-DSS operations, and which network types (e.g., 5G/NR, 4G/LTE) are to be used by the network operator **54**. The spectrum grant request may also include a low frequency and/or a high frequency of a preferred frequency band and/or an identifier for the base station **50B** and/or the network operator **54**. The identifier may uniquely identify the base station **50B** and distinguish the base station **50B** from the other base stations **50**, and/or may include an indication of the network operator **54**. Data included in the spectrum grant request may represent a variety of data types, and thus may include data types of strings, number, objects, Boolean values, NULL values, or the like.

(68) At block **168**, the SAS entity **56** confirms association of the base station **50B** with the network operator **54**. In particular, the spectrum grant request may include an authentication parameter that enables the SAS entity **56** to confirm the association of the base station **50B** with the network operator **54**. This may involve verifying that an identifier of the base station **50B** corresponds to a list of known associations between base stations **50** and the network operator **54**.

(69) At block **170**, the SAS entity **56** determines a channel allocation assignment for the network operator **54** corresponding to the base station **50B**. To do so, the SAS entity **56** may query its memory to determine whether a channel allocation assignment has already been provided to the network operator **54**, and, if so, may report the previously reported channel allocation assignment to the base station **50B** if the channel allocation assignment remains active and/or valid (e.g., has not yet timed out or been invalidated due to a change in operation or cumulative network demand from each network operator **54**). The SAS entity **56** may determine the channel allocation assignment as a channel that satisfies a maximum number of criteria or parameters included in the spectrum grant request. The SAS entity **56** may, in some cases, assign more than one channel to a network operator **54**. The multiple channel assignments may be in response to a parameter included in the spectrum grant request, in response to stored rule defining a particular parameter to be followed or satisfied when assigning a channel to the network operator **54**, in response to the SAS entity **56** receiving multiple spectrum grant requests from different base stations **50** of the network operator **54**, or the like.

(70) Indeed, to do so, the SAS entity **56** may test out different channels and determine a number of preferences met by the hypothetical assignment to the network operator **54** before determining which hypothetical assignment yields the relatively better outcome for the network operator **54** (e.g., matches the highest number of preferences). Indeed, the SAS entity **56** may use any suitable algorithm to meet a maximum or relatively high number of criteria or parameters in the spectrum

grant request. This may also involve the SAS entity **56** resolving any higher priority spectrum grant requests prior to resolving the spectrum grant request of the base station **50B**. It is also noted, however, that in some embodiments, the SAS entity **56** may resolve one or more spectrum grant requests by determining a relatively best-case scenario that satisfies a higher number of preferences requested by a variety of network operators **54**.

(71) For example, the SAS entity **56** may identify unallocated channels corresponding to the raster requirements for the different networks of multiple spectrum grant requests. The SAS entity **56** may compare each preference of the multiple network operators **54** requesting assignment of an unallocated channel. The SAS entity **56** may resolve as many preferences with hypothetical assignments to obtain a preference-granted metric (e.g., percentage value indicating a total number of network operator **54** preferences satisfied divided by a total number of network operator **54** preferences). Then, once a threshold number of hypothetical assignments are generated, the SAS entity **56** may enact the hypothetical assignment corresponding to the relatively highest preference-granted metric. When there is a tie between two or more hypothetical assignments (e.g., where a network operator **54** is not able to have 100 percent (%) of preferences filled), a final decision may be based on granting the higher priority network operators **54** a channel that satisfies relatively higher amounts of their preferences and/or meets 100% of their preferences when compared to meeting preferences of lower priority network operators **54**. This determination may involve the SAS entity **56** acting upon the “spectrum usage” parameter (e.g., UsageType) from the spectrum grant request and allocating spectrum on the corresponding raster that is compatible with the spectrum usage parameter. Furthermore, in some embodiments, the SAS entity **56** may prioritize certain parameters, such as either based on its own prioritization, input from one of the base stations **50**, or both, and may determine the channel assignment based on meeting higher priority parameters before meeting lower priority parameters.

(72) At block **172**, the SAS entity **56** transmits the channel allocation assignment for the network operator **54** to the base station **50B** and, at block **174**, stores the channel allocation assignment in memory as associated with the network operator **54** and/or base station **50B**. That is, the SAS entity **56** may store an indication of the channel allocation assignment in memory as associated with the identifier for the base station **50B** and/or the network operator **54** (received at block **166**).

(73) To elaborate further on operations that the SAS entity **56** may perform to determine the channel allocation assignment, FIG. **13** is a flowchart of a method **190** for operating the SAS entity **56** to determine a channel to assign to the base station **50B** when performing operations of block **170** of the method **164** of FIG. **12**, according to embodiments of the present disclosure. It is noted that, although depicted in a particular order, the blocks of the method **190** may be performed in any suitable order, and at least some blocks may be skipped altogether. As described herein, the method **190** is described as performed by the SAS entity **56**, however, it should be understood that any suitable processing and/or control circuitry may perform some or all of the operations of the method **190**, such as an electronic device **10** via one or more of the processors **12**.

(74) The SAS entity **56** may determine of a channel allocation assignment for a network operator **54** at block **170** of FIG. **12**. When performing operations of block **170** of FIG. **12**, the SAS entity **56** may determine whether the spectrum grant request includes indication of the network operator **54** using DSS operations. These determinations may include performance of operations of the method **190** of FIG. **13**. Indeed, after resolving any high priority requests (e.g., requests for a channel assignment from a network operator **54** associated with a relatively higher priority), the SAS entity **56** may proceed from block **170** of FIG. **12** to determine, at block **194**, whether the spectrum grant request corresponds to a DSS-related request (e.g., a request for a channel usable or compatible with DSS operations). In some embodiments, however, it is noted that the SAS entity **56** may assign channel allocations without first resolving priorities of requests, using the priorities of the network operator **54** merely to break ties and/or to determine final assignments in the event that a compromising arrangement is unable to be determined.

(75) Indeed, at block **194**, the SAS entity **56** determines whether the spectrum grant request corresponds to a DSS-related request. To do so, the SAS entity **56** may refer to the usage type parameter (e.g., “spectrum usage” parameter, UsageType). The usage type parameter may specify, for a two-network options case (e.g., 4G/LTE vs 5G/NR), whether the network operator **54** is requesting a channel compatible with a first network raster, with a second network raster, or both. For example, the usage type parameter may identify “LTE,” “NR,” or “LTE/NR DSS” (e.g., spectrum sharing flag). It is noted that other network option cases may be used, such as three or more network options, where each network may be associated with different frequency and/or raster configurations.

(76) In response to determining that the usage type parameter corresponds to a DSS operation, at block **196**, the SAS entity **56** determines a channel allocation assignment that is compatible with a raster of a first network and of a second network. For example, certain specifications may define that channels used for 4G/LTE **48** (e.g., n48) band communications are to align with a raster of 100 kilohertz (kHz) while channels used for 5G/NR **48** (e.g., n48) band communications are to align with a raster of 15 kHz or 30 kHz. Thus, when the usage type parameter indicates “LTE/NR DSS,” the SAS entity **56** may assign a channel that aligns with both the 4G/LTE raster and the 5G/NR raster, such as 300 kHz. For example, referring to channel **94** of FIG. **9A**, the SAS entity **56** may determine that the first operator **92A** indicated a preference to use DSS operations, and thus assigns the first operator **92A** the channel **94** having a center frequency of 3600 MHz, which is a multiple of the 300 kHz raster.

(77) If the SAS entity **56** determines that the usage type parameter does not correspond to a DSS operation, then, at block **198**, the SAS entity **56** may determine whether the spectrum grant request corresponds to a first network related request, such as whether the usage type parameter indicates “LTE.” When the SAS entity **56** determines that the spectrum grant request corresponds to the first network related request, at block **200**, the SAS entity **56** determines a channel allocation assignment that is compatible with a raster of the first network without concern for whether the channel is compatible with a raster of the second network. For example, the SAS entity **56** may determine the channel to be a channel that aligns with a raster of 100 kHz used for LTE communications. For example, referring to channel **102** of FIG. **9A**, the SAS entity **56** may determine that the fifth operator **92E** indicated a preference to use LTE operations but not use DSS operations, and thus assigns the fifth operator **92E** the channel **102** having a center frequency of 3625 MHz, which is a multiple of 100 kHz (but not 15 kHz or 30 kHz, e.g., the rasters that may be used for NR communications).

(78) However, when, at block **198**, the SAS entity **56** determines that the spectrum grant request does not correspond to the first network related request, the SAS entity **56** may default to determining that the spectrum grant request corresponds to a second network related request and/or may determine that the usage type parameter indicates “NR.” As such, when the SAS entity **56** determines that the spectrum grant request corresponds to the second network related request, at block **202**, the SAS entity **56** determines a channel allocation assignment that is compatible with a raster of the second network without concern for whether the channel is compatible with a raster of the first network. For example, the SAS entity **56** may determine the channel to be a channel that aligns with a raster of 15 kHz or 30 kHz used for NR communications. After determining the channel allocation assignment, the SAS entity **56** may proceed to perform operations of block **172** of FIG. **12** (e.g., to perform remaining operations of the method **164**).

(79) The method **190** represents one example method of determining the channel allocation assignment for the network provided. Other methods may also be used. For example, in some cases, the base station **50B** may not include the usage type parameter in the spectrum grant request. In these cases, the SAS entity **56** may reference rules that indicate operational preferences and/or expectations for each network operator **54**. The rules may be stored in a memory (or storage) of the SAS entity **56**. In some cases, the rules may be accessible by the SAS entity **56** in a cloud-based

storage system also accessible by one or more network operators **54**, where the network operators **54** may respectively update communication preferences by updating a definition of a rule in the cloud-based storage system. In this way, a network operator **54** may specify in a rule a preference for implementing DSS operations, and thus the SAS entity **56** may access the rule in memory that indicates the preference when determining a channel allocation assignment for that network operator **54**. Keeping this in mind, the methods of FIGS. **10-13** may be altered to accommodate for the use of these rules (e.g., the SAS entity **56** may refer to the rules rather than relying on the usage parameter of the spectrum grant request as transmitted in operations of block **146** of FIG. **11** and received in operations of block **166** of FIG. **12**).

(80) Indeed, in some cases, the spectrum grant request may include several parameters that identify the network operator **54** and the base station **50B**, such as a user identifier, a governmental identifier corresponding to the network operator **54** (e.g., a Federal Communications Commission identifier (FCC ID)), a serial number and/or identifier of the base station **50B**, and/or the like. The rules discussed above may be accessible and searchable by the SAS entity **56** based at least in part on one or more of these identifiers that may be included in the spectrum grant request. Thus, when a particular network operator **54** is interested in defaulting to performing DSS operations (or defaulting to being assigned a channel compatible with performance of DSS operations), the SAS entity **56** may have a rule formed and/or negotiated with the network operator **54** to reference when assigning channels. In this way, when a request is received corresponding to the network operator **54**, the SAS entity **56** determines to assign the network operator **54** a channel compatible with DSS operations based on accessing the definition of the rule for the network operator **54**. For example, a rule may define that when the first network operator **92A** transmits a spectrum grant request to the SAS entity **56**, the SAS entity **56** is to default to assigning a DSS operation compatible raster (e.g., include a definition that associates the first network operator **92A** to DSS operations for purposes of channel assignment). When defining a rule, the network operator **54** and/or the SAS entity **56** may specify a default action that is to take place when assigning a channel to the network operator **54** in response to the SAS entity **56** identifying a parameter of the spectrum grant request. For example, the SAS entity **56** may determine the spectrum grant request to include an identifier corresponding to the network operator **54**, and in response to this determination, access a rule defining a default action to assign to the network operator **54** DSS operation-compatible channel or a non-DSS operation compatible channel. Other default actions may involve a default network deployment, default channel bandwidth to be assigned, default number of channels to be assigned to the network operator **54**, or the like, or any combination thereof.

(81) The SAS entity **56** may update the rules and definitions indicated by the rules over time, such as when requested by the network operator **54** to update the rule. Indeed, the network operator **54** may initiate the change in the rule from corresponding an identifier for the network operator **54** to a first default action to corresponding the identifier for the network operator **54** to a second default action. In this way, the rule may include a first definition at a first time and a second definition at a second time and where the network operator **54** is able to negotiate or notify with the spectrum access system what definition is stored as the rule. Furthermore, in some cases, there may be communication parameters defined by a standards body, such that when a particular identifier, flag is set, condition is met, a signal is transmitted on a frequency or with a certain amplitude, or the like, the SAS entity **56** knows to assign to a channel compatible with DSS operations to a network operator **54** corresponding to the particular identifier, flag, condition being met, or the like.

(82) It is noted that although described in terms as a single SAS entity **56**, one or more SAS entities may perform the operations described herein. When multiple SAS entities manage channel allocations for a particular geographical region, each SAS entity **56** may benefit from knowing when a channel is no longer available for assignment (e.g., previously allocated channel, unavailable for allocation). When multiple SAS entities are assigning channel allocations to different network operators **54** and/or base stations **50**, the respective SAS entities may

intercommunicate information regarding allocated frequency ranges to reduce, minimize, or eliminate a likelihood of different network operators **54** being assigned a same channel. This communication may be a report that is periodically sent between SAS entities to update each other regarding assignment changes of the channels. In some cases, however, the updates to channel assignments may be transmitted to each of the SAS entities after a channel assignment is determined. By receiving these reports, each of the SAS entities may assign unallocated channels (e.g., unallocated frequency ranges) without inadvertently assigning an allocated channel (e.g., allocated frequency range).

(83) In some cases, when the base station **50B** receives the spectrum allocation assignment, the base station **50B** may continue on to perform dynamic spectrum sharing (DSS) operations. DSS operations may include the base station **50B** determining a current network load (e.g., a first network load) and determining whether the determined current network load is greater than a load threshold. Indeed, the DSS operations may be used to change a network used by the base station **50B** to communicate with user equipment, such as in response to changes in network loads (e.g., network demands). For example, the 4G/LTE network may use less bandwidth to downlink a message to a user equipment than 5G/NR, and thus may be more suitable to use when network loads are greater than the load threshold. In this way, in response to determining the current network load is greater than or equal to the load threshold, the base station **50B** may transmit a message to the user equipment using the 4G/LTE network (e.g., first network), and in response to determining the current network load is less than the load threshold, transmitting the message to the user equipment using the 5G/NR (e.g., second network). In some cases, the base station **50B**, using DSS operations, communicates with a first user equipment using 4G/LTE and with a second user equipment using 5G/NR at least partially simultaneously and without regard for network load.

(84) Technical effects of the present disclosure include systems and methods for improving allocation of resources from a shared pool of network resources by a SAS entity. Indeed, these systems and methods provide for ways for a network operator to communicate with a SAS entity, via a base station, one or more preferences to be considered by the SAS entity when assignment communication channels of a radio spectrum. Each network operator may indicate to the SAS entity whether the respective network operator expects to use a first network, a second network, or DSS operations that use both when communicating with user equipment in its serving cells. When the SAS entity is determining a channel to assign a network operator that has indicated the expected use of DSS operations, the SAS entity may determine a channel that has a center frequency which aligns with a raster compatible with raster definitions for both the first network and the second network. However, when the SAS entity is determining a channel to assign a network operator that has indicated just one network is to be used for communication, the SAS entity may determine a channel that has a center frequency which aligns with a raster compatible with the one network. Relatively defined priorities of the network operators may also be considered by the SAS entity when determining to assign a particular channel.

(85) The specific embodiments described above have been shown by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.

(86) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A base station, comprising: a processor; and a memory comprising machine-readable instructions that, when executed by the processor, cause the processor to: receive a registration request from a user device; transmit an indication of the registration request to a network operator to register the user device with the network operator; transmit a request for a spectrum allocation to a spectrum access system, the request including a usage parameter indicating that the network operator is to communicate using a spectrum sharing operation, the usage parameter comprising a spectrum sharing flag; receive a channel allocation assignment from the spectrum access system, the spectrum access system being configured to generate the channel allocation assignment based at least in part on the usage parameter, the channel allocation assignment defining a range of frequencies used when sending communications to the user device; and communicate with the user device using the spectrum sharing operation, the channel allocation assignment, and the range of frequencies.
2. The base station of claim 1, wherein the range of frequencies comprises frequencies between 3550 megahertz (MHz) and 3700 MHz.
3. The base station of claim 1, wherein the range of frequencies is associated with a first raster of 100 kilohertz (kHz) and with a second raster of 15 kHz or 30 kHz.
4. The base station of claim 1, wherein the network operator is configured to communicate with another user device using frequencies associated with a raster of 100 kilohertz (kHz) while communicating with the user device.
5. The base station of claim 1, wherein the network operator is configured to communicate with another user device using frequencies associated with a raster of 15 kilohertz (kHz) or of 30 kHz while communicating with the user device.
6. The base station of claim 1, wherein the network operator is configured to generate the channel allocation assignment based at least in part on a determination of unallocated frequency ranges.
7. The base station of claim 1, wherein the spectrum sharing operation comprises a dynamic spectrum sharing (DSS) operation that enables the network operator, using the range of frequencies, to communicate with the user device on a first network and with another user device on a second network.
8. The base station of claim 1, wherein the indication of the registration request comprises a subscriber identification module (SIM) card identifier.
9. A base station, comprising: a transmitter configured to transmit a request for a spectrum allocation to a spectrum access system, the request including an identifier indicating a network operator and a usage parameter indicating that the network operator is to communicate using a spectrum sharing operation, the usage parameter comprising a spectrum sharing flag; a receiver configured to receive a channel allocation assignment from the spectrum access system based at least in part on the identifier, the channel allocation assignment defining a range of frequencies used when sending communications to a user device based on a rule associated with the identifier and stored in memory, the rule indicating that the network operator is to communicate using the spectrum sharing operation; and one or more processors configured to communicate with the user device via the transmitter and the receiver using the spectrum sharing operation, the channel allocation assignment, and the range of frequencies.
10. The base station of claim 9, wherein the network operator is configured to change the rule to indicate that the network operator is to communicate using a non-spectrum sharing operation, causing the one or more processors to communicate with the user device using the non-spectrum sharing operation.
11. The base station of claim 9, wherein the network operator is configured to change the rule to indicate that the network operator is to communicate using a first network and a second network as

part of the spectrum sharing operation, causing the one or more processors to communicate with the user device using the first network or the second network.

12. The base station of claim 9, wherein the spectrum sharing operation involves a first network corresponding to a raster of 100 kilohertz (kHz) and a second network corresponding to a raster of 15 kHz or 30 kHz.

13. The base station of claim 12, wherein the channel allocation assignment corresponds to a raster of 300 kilohertz.

14. The base station of claim 9, wherein the spectrum sharing flag indicates a respective network type of a plurality of network types.

15. A method, comprising: transmitting a request for a spectrum allocation to a spectrum access system, the request including a usage parameter indicating that a network operator is to communicate with one or more user equipment using a first network and using a second network, the usage parameter comprising a spectrum sharing flag; receiving a channel allocation assignment from the spectrum access system, the spectrum access system being configured to generate the channel allocation assignment based at least in part on the usage parameter, the channel allocation assignment defining a range of frequencies used when sending communications to the one or more user equipment; and communicating with the one or more user equipment using a spectrum sharing operation, the channel allocation assignment, and the range of frequencies.

16. The method of claim 15, comprising: performing dynamic spectrum sharing operations by determining a network load of the second network; determining whether the network load is greater than or equal to a load threshold; in response to determining that the network load is greater than or equal to the load threshold, transmitting a message to the one or more user equipment using the first network; and in response to determining that the network load is less than the load threshold, transmitting the message to the one or more user equipment using the second network.

17. The method of claim 15, comprising: performing dynamic spectrum sharing operations at least in part by: assigning a first sub-channel aligned with a 100 kilohertz (kHz) raster to a first user equipment of the one or more user equipment; assigning a second sub-channel aligned with a 15 kHz raster or a 30 kHz raster to a second user equipment of the one or more user equipment; and communicating with the first user equipment using the first sub-channel while communicating with the second user equipment using the second sub-channel.

18. The method of claim 15, comprising: assigning to a first user equipment of the one or more user equipment a first sub-channel of the range of frequencies and assigning to a second user equipment of the one or more user equipment a second sub-channel of the range of frequencies, wherein the range of frequencies comprises a center frequency aligned on a 300 kilohertz (kHz) raster; transmitting a first message via the first sub-channel to the first user equipment, wherein the first sub-channel is used for communicating via the first network; and transmitting a second message via the second sub-channel to the second user equipment, wherein the second sub-channel is used for communicating via the second network.

19. The method of claim 18, comprising: pausing transmission of a third message to the first user equipment based at least in part on a communication configuration corresponding to the first user equipment; and receiving a fourth message via the first sub-channel from the first user equipment.

20. The method of claim 19, comprising: determining an upcoming downlink allocation to start at a first time based at least in part on the communication configuration corresponding to the first user equipment; and transmitting the third message at the first time.
